Experiment 21
The oscilloscope

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Abstract

This experiment explored the use of a digital oscilloscope to measure and analyze electrical signals. Basic waveform measurements were performed in X–T mode using sinusoidal, square, and sawtooth signals, verifying the oscilloscope's ability to accurately display amplitude, frequency, and phase. An RC circuit was then studied to determine the characteristic frequency through both phase shift and voltage ratio methods, yielding results consistent with theoretical predictions. Finally, Lissajous figures were used in X–Y mode to visualize phase differences between signals. The experiment confirmed the oscilloscope's effectiveness as a tool for signal analysis and reinforced key concepts in circuit behavior.

Contents

1 Introduction **2**
1.1 Contextualization . . . 2
1.2 Theorical principles . . . 2
1.2.1 Operation of Analog and Digital Oscilloscopes . . . 2
1.2.2 X-T mode . . . 3
1.2.3 X-Y mode . . . 4

2 Materials and methods **5**
2.1 Materials . . . 5
2.2 Methods . . . 6
2.2.1 measurements in X-T mode . . . 6
2.2.2 Series RC circuit . . . 6
2.2.3 Lissajous figures . . . 7

3 Results and discussion **8**
3.1 Basic measurements in X-T mode . . . 8
3.2 Series RC circuit . . . 10
3.3 Lissajous figures . . . 13

4 Conclusions **14**

5 Appendixes **14**
5.1 Calculation of Uncertainties . . . 14

1 Introduction

1.1 Contextualization

Oscilloscopes are fundamental instruments in the observation and analysis of electrical signals. Their primary function is to graph the voltage as a function of time, allowing users to study waveform characteristics such as amplitude, frequency, rise time, distortion, and phase difference.

The history of oscilloscopes began in the late 19th century with the development of the cathode-ray tube (CRT). The first CRT-based devices were used to display simple electronic signals, and by 1930 s, the first true oscilloscopes were introduced, notably by Allen B. DuMont [1]. The use of triggered sweep mechanisms in 1940 s and 1950 s allowed engineers to stabilize and study non-repeating signals, revolutionizing electronic diagnostics.

The transition from analog to digital oscilloscopes began in the 1980s, integrating digital storage and signal processing capabilities that vastly expanded their analytical power.

Today, digital oscilloscopes are used in a wide array of industries. In automotive engineering, they are used to analyze communication protocols; and, in telecommunications, they help in the development and maintenance of high-speed data networks; while in the medical field, oscilloscopes contribute to the analysis of electrophysiological signals.

That way, we can say their capacity to capture, store, and analyze periodic and transient events makes them indispensable in both research and industrial settings.

1.2 Theorical principles

1.2.1 Operation of Analog and Digital Oscilloscopes

Traditional analog oscilloscopes use a CRT and deflection plates: the input signal drives a vertical amplifier that deflects the electron beam in proportion to voltage, while a time-base generator sweeps the beam horizontally. This controlled deflection is what allows the visualization of signals on the fluorescent screen.

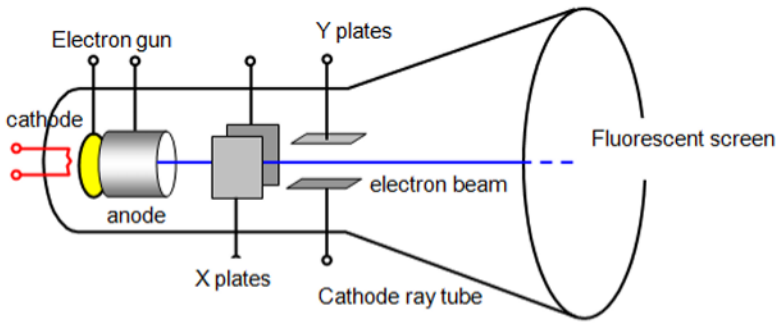


Figure 1: Analog oscilloscope's system [2]

This direct tracing requires carefully synchronized triggering to produce a stable waveform image: There are amplifiers that are connected to the deflection plates. In the most common operating mode, the waveform we analyze is fed into the vertical input terminals and the vertical amplifier boosts the signal's amplitude to produce a noticeable vertical displacement of the electron beam. Meanwhile, the horizontal deflection is achieved using a sweep generator, often

called the "time base." Its purpose is to generate sawtooth voltages that move the beam horizontally. If the period of the vertical signal matches that of the time base signal, the waveform appears stationary on the screen (Figure 2). To maintain a stable display, the time base must be triggered in sync with the input signal.

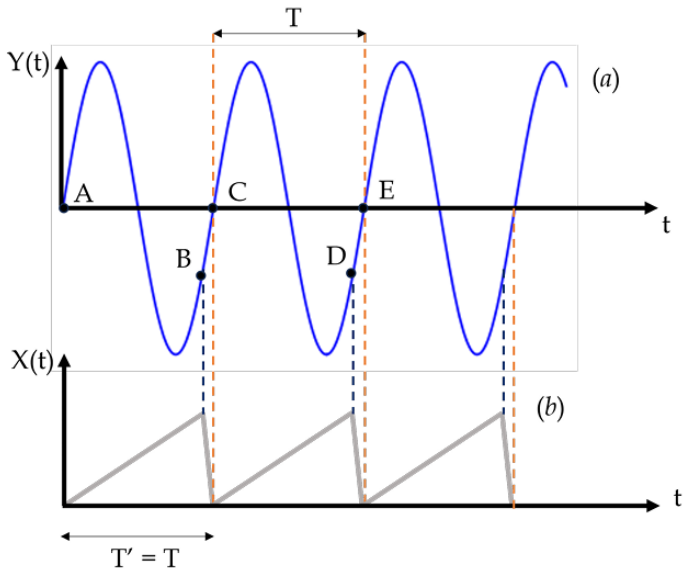


Figure 2: Fixed image created by Horizontal and Vertical signals [2].

However, analog oscilloscopes have significant limitations such as they can not easily capture one-shot or very fast transient events. In contrast, modern digital oscilloscopes overcome these limitations by digitizing the input and storing waveforms for display. Unlike analog CRT scopes, which simply deflect a beam directly, digital scopes use internal memory and processors to render the waveform, allowing features like persistent display, automatic measurements, and complex triggering.

That way, modern digital oscilloscopes sample the input signal at high speed and store data in memory. They then reconstruct the waveform for display. This digitization allows powerful features (for example long memory traces, digital filtering, automated measurements, and capturing non-repetitive events that analog scopes can not retain). In practice, nearly all new oscilloscopes are digital, combining the fundamental tracing ability of the analog front end with versatile digital signal processing.

The digital oscilloscope used in this lab is a dual-channel instrument, meaning it can sample and display two independent signals simultaneously.

1.2.2 X-T mode

In its standard X–T mode (time mode), each channel displays voltage versus time on the screen, allowing direct measurement of waveform amplitude, period, frequency, and phase. The user can scale the horizontal (time) axis and vertical axes independently for each channel, being able to view wave shapes, amplitudes, and time relationships simultaneously.

For sinusoidal signals of equal frequency f with a phase shift ϕ we know the genereal expres-

sion

$$V_a(t) = V_a cos(2\pi f t) \tag{1}$$

$$V_b(t) = V_b cos(2\pi f t + \phi) \tag{2}$$

where Va and Vb are the amplitudes of the respective signals, and ϕ represents the phase difference between them.

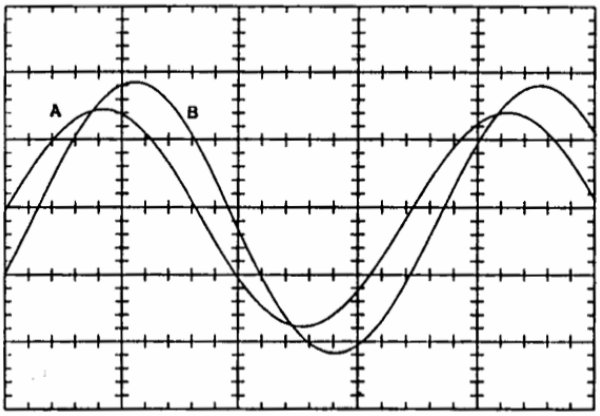


Figure 3: Screen in the X-T mode, where two sinusoidal signals of equal frequency f are input with a phase difference of ϕ [2].

1.2.3 X-Y mode

In contrast, X–Y mode uses one channel to control horizontal deflection and the other for vertical deflection. The oscilloscope then plots one signal versus the other, producing a parametric graph.

To goo deeper, we will consider two sinusoidal signals of same frequency with a certain phase shift, ϕ:

$$x(t) = X cos(\omega t) \tag{3}$$

$$y(t) = Y cos(\omega t + \phi) \tag{4}$$

If the points $(x(t), y(t))$ are represented, for each time, on a plane, an ellipse will be obtained, which may become a straight line or a circle. This can be obtained by selecting the X-Y mode on the oscilloscope, to do so, each signal should be connected to a channel, and then the 'Acquire' button should be pressed in function button area, and X-Y mode can be turned on with the menu selection buttons.

Done this, a graph will appear at the bottom of the screen, representing the Y vs. X signal (CH2 vs. CH1). With that, the user is able to measure the phase shift between both signals, which should satisfy

$$\phi = arcosen\left(\frac{OC}{OD}\right) = arcosen\left(\frac{CC'}{DD'}\right) \tag{5}$$

The points C, C', D, and D' are easily measurable allowing the determination of the phase shift. Although this calculation can be performed by visually inspecting the ellipse on the screen, the digital oscilloscope automatically does it. Therefore, at the bottom right, some measurements will appear: T (period), F (frequency), V (baseline voltage), Vpp (peak-to-peak voltage), and RP (relative phase between the two signals).

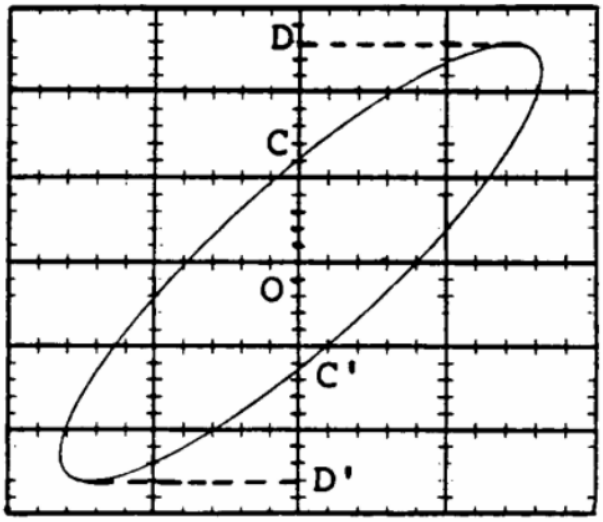


Figure 4: Screen in the X-Y mode, where two sinusoidal signals are input with equal frequency f and a phase shift of ϕ [2].

2 Materials and methods

2.1 Materials

The materials used in this practice are:

- Oscilloscope: a sophisticated instrument commonly employed for tasks like circuit diagnostics and analog signal analysis, as it visually displays voltage changes over time. To operate it, the device must be connected in parallel across the signal's measurement points.
- Capacitors: two-terminal components capable of storing energy as an electric charge. They consist of two conductive plates separated by a small gap, which may be a vacuum or a dielectric insulating material. The capacitor used in the experiment had a 4.7 microfarad (μF) capacitance.
- Breadboard: a small, reusable platform for prototyping electronic circuits without soldering. Components are easily interconnected by inserting their leads into the board's pre-aligned holes.
- Function Generator: an electronic tool designed to produce electrical currents with adjustable properties, including frequency and amplitude. It can generate various waveform types, such as sine waves or square waves (figure 5).
- Resistors: passive components that regulate current flow within an electrical circuit. In our case, with a value of 100Ω.
- BNC cables and adapters: for making reliable connections between the generator, oscilloscope channels, and circuit.



Figure 5: Function generator

2.2 Methods

2.2.1 measurements in X-T mode

First, we wil connect the function generator's channel A output (CHA) to channel 1 (CH1) of the oscilloscope using a BNC cable. On the function generator, we will select a sine waveform at a convenient frequency and amplitude.
On the oscilloscope, the "Auto-Set" button will be pressed so that the waveform is displayed stably. After that, we will be able to record the measured frequency and peak-to-peak voltage from the oscilloscope readouts, and compare to the generator settings.
This process will be repated for each waveform type of interest (sinusoidal, quadratic and saw-tooth, in our case) and for various amplitudes or frequencies, noting any differences between the generated signal and the oscilloscope measurement.

2.2.2 Series RC circuit

On the breadboard, we will assemble the series circuit by connecting the 100Ω resistor in series with the $4.7\mu F$ capacitor, with the far end of the capacitor tied to ground. Then, we will connect the function generator output to the input of the series chain and use a BNC T-adapter so that the generator output feeds both the resistor and oscilloscope channel 1 (CH1) simultaneously.

Specifically, one T-adapter branch will connect to the series RC, and the other branch will do it directly to CH1 of the scope. Then, we will connect CH2 of the scope to the junction between the resistor and capacitor (or equivalently across the capacitor). This way, CH1 will measure the input voltage V_e, while CH2 will measure the capacitor voltage V_C.

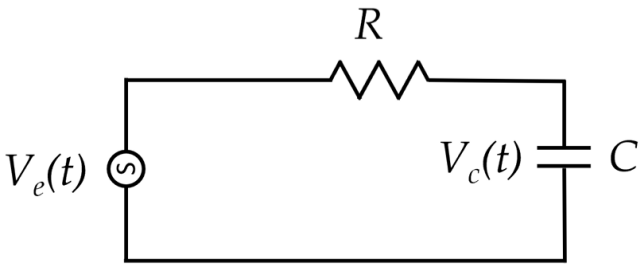


Figure 6: Circuit with V_e and V_C represented [2].

From that, once the circuit in the figure 6 has been built, applying a sinusoidal source (starting at low frequency) we will observe both traces in X–T mode and slowly increase the frequency in steps so we will be able to record record the amplitude of V_C and the phase difference (wich should go from 0° to little over 90°) between CH1 and CH2 at each step.

Theoretically, if the input voltage V_e is sinusoidal and given by

$$V_e(t) = V_e cos(2\pi ft), \tag{6}$$

the voltage across the capacitor will also be sinusoidal but with a phase shift. Its expression will be

$$V_C(t) = V_c cos(2\pi ft + \phi) \tag{7}$$

where the amplitude V_c is determined by:

$$V_c = \frac{V_e}{\sqrt{1 + (2\pi ft)^2 R^2 C^2}} \tag{8}$$

and the phase shift ϕ between the input signal and the voltage at the capacitor

$$\phi = -arctan(2\pi fRC), \tag{9}$$

where R and C are the values of the resistor and the capacitor, which we will assume to be exact values, with no uncertainties.

Lastly, we can determinate the natural frequency of resonance of the RC circuit, known as the characteristic frequency f_{char} with the expression

$$f_{char} = \frac{1}{2\pi RC} \tag{10}$$

and the relation V_C/V_e:

$$\frac{V_c}{V_e} = \frac{1}{\sqrt{1 + (f/f_{char})^2}} \tag{11}$$

2.2.3 Lissajous figures

Lissajous figures are formed when two harmonic motions intersect at a single point while moving in perpendicular directions. These oscillations may share identical frequencies and phases, or they may differ in one or both aspects.

To observe this phenomenon using an oscilloscope, there will be required two sinusoidal voltage signals (each connected to a separate input channel). One signal will serve as a reference while for the other we will operate using the function of the oscilloscope called "X-Y Mode" This

configuration will enable us to measure the phase differences between two signals of identical frequency.

When the input signals have different frequencies, the resulting patterns on the oscilloscope display may appear as either closed or open shapes, depending on the specific frequency ratio between them as shown in figure 7

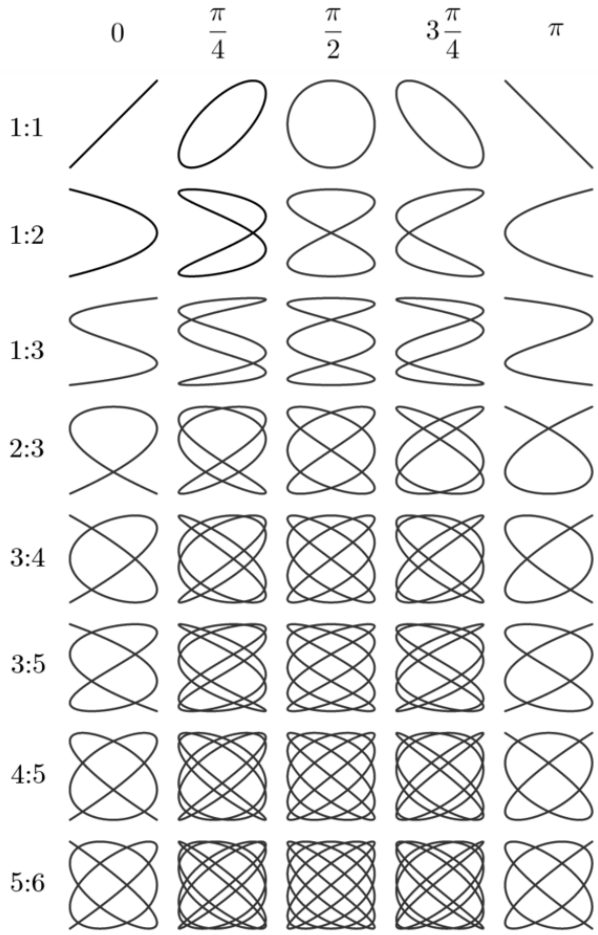


Figure 7: Lissajous figures [3].

3 Results and discussion

3.1 Basic measurements in X-T mode

In this first part of the experiment we measured the frequency and voltage with the oscilloscope, comparing them with the output values of the generator.

Sinusoidal, square and sawtooth waves were used. The results are organised in tables 1, 2 and 3 respectively. In addition, figures 8, 9 and 10 are an example of the images displayed by the oscilloscope for said signals.

	Oscilloscope data				Generator data	
	$f\ (kHz)$	$u_C(f)\ (kHz)$	$V\ (Vpp)$	$u_C(V)\ Vpp$	$f\ (kHz)$	$V\ (Vpp)$
1	10,000	0,011	5,0000	0,2002	10	5
2	240,10	0,29	2,98	0,11	240	3
3	29990,00	1441,56	9,28	0,38	30000	9,5

Table 1: Frequency and voltage measured by the oscilloscope, with their corresponding errors; and output values of the power supply of this parameters for the sinusoidal wave.

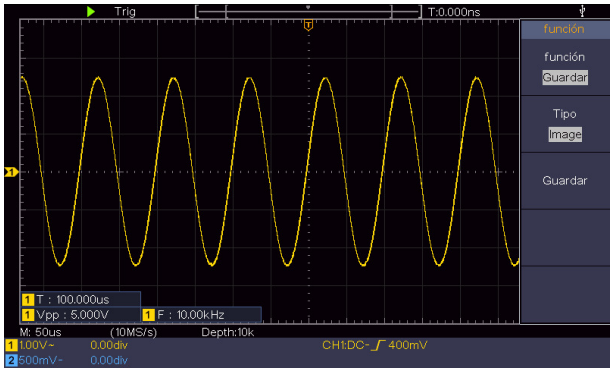


Figure 8: Screenshot of the sinusoidal wave displayed by the oscilloscope.

	Oscilloscope data				Generator data	
	$f\ (Hz)$	$u_C(f)\ (Hz)$	$V\ (Vpp)$	$u_C(V)\ Vpp$	$f\ (Hz)$	$V\ (Vpp)$
4	1000,0	1,1	9,52	0,39	1000	9,5
5	35970	30	4,00	0,17	36000	4
6	50000	57	2,020	0,086	50000	2

Table 2: Frequency and voltage measured by the oscilloscope, with their corresponding errors; and output values of the power supply of this parameters for the squared wave.

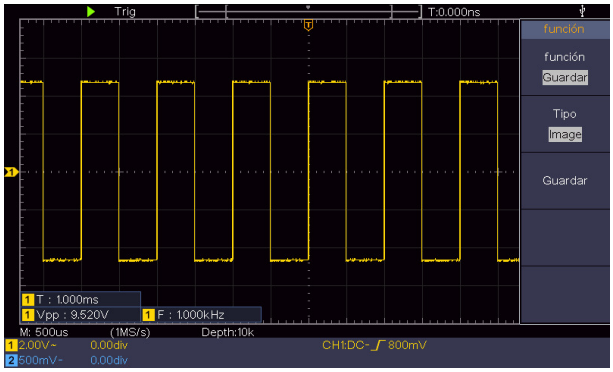


Figure 9: Screenshot of the squared wave displayed by the oscilloscope.

	Oscilloscope data				Generator data	
	$f\ (kHz)$	$u_C(f)\ (kHz)$	$V\ (Vpp)$	$u_C(V)\ Vpp$	$f\ (kHz)$	$V\ (Vpp)$
1	50,000	0,057	0,980	0,055	50	2
2	84,990	0,085	0,616	0,023	85	1
3	7,0030	0,0056	4,04	0,17	7	8

Table 3: Frequency and voltage measured by the oscilloscope, with their corresponding errors; and output values of the power supply of this parameters for the sawtooth wave.

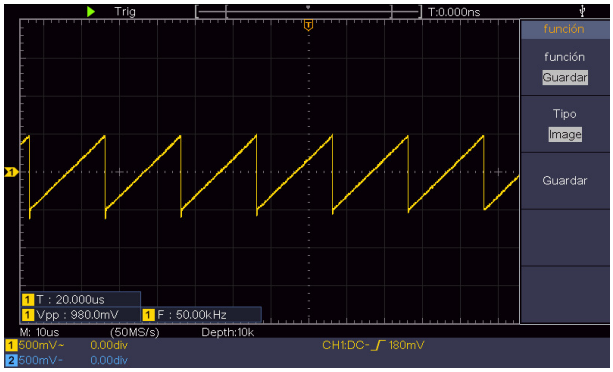


Figure 10: Screenshot of the sawtooth wave displayed by the oscilloscope.

As we can see in tables 1 and 2, the values measured with the oscilloscope are quite similar to the output values of the generator. What is more, the uncertainties of both parameters are very low, which reinforces the veracity of our results.

Regarding the sawtooth signal, there is a clear agreement between the two values obtained for f. However, the voltage data differ greatly. In fact, the number measured with the oscilloscope is half the output value. We can attribute this error to instrumentation inaccuracy. It is likely that the problem is the oscilloscope - which may not be measuring well the voltage supplied for the sawtooth wave -, since when other signals were used the results were satisfactory.

3.2 Series RC circuit

A resistor of 100 Ω and a $4,7\ \mu F$ capacitor were employed to construct the circuit sketched in figure 6. With these values it is possible to obtain the characteristic frequency of the circuit using equation 14. Then, f_{char} is equal to $f_{char} = 338,63\ Hz$.

The following table shows the phase shift between output signal from the generator (V_e) and the signal of the capacitor (V_c); as well as the ratio V_c/V_e and their corresponding uncertainties. These values were obtained changing the frequency of the signal from 10 to 10000 Hz, following a logarithmic function.

$f\ (Hz)$	$u_C(f)\ (Hz)$	$\phi\ (°)$	$u_\phi\ (°)$	V_c/V_e	$u_C(V_c/V_e)$
10,000	0,011	14,40000	0,00029	1,000	0,057
20,000	0,018	8,62600	0,00029	0,962	0,074
40,000	0,036	14,40000	0,00029	0,980	0,076
70,030	0,056	12,60500	0,00029	0,942	0,073
100,00	0,11	18,00000	0,00029	0,900	0,072
200,00	0,18	28,80000	0,00029	0,889	0,075
400,00	0,36	48,96000	0,00029	0,650	0,051
700,30	0,56	60,50400	0,00029	0,46	0,029
1000,0	1,1	72,00000	0,00029	0,349	0,024
2000,0	1,8	77,76000	0,00029	0,182	0,012
4000,0	3,6	83,52000	0,00029	0,0917	0,0059
7005	5,6	88,11200	0,00029	0,0530	0,0033
10000	11	90,00000	0,00029	0,038	0,0026

Table 4: Frequency f, lag ϕ and ratio between signals V_c and V_e, with their respective errors.

It is noticeable that the ratio tends to zero, as we were expecting. Furthermore, the combined uncertainties calculated are small, meaning that our results are reliable.

With this data, we were able to plot two logarithmic graphics. The first image (11) represents the phase shift ϕ over frequency f. Whereas the second one (12) portrays the relation between the ratio V_c/V_e and f. The information obtained from these graphs are organized in tables 5 and 6.

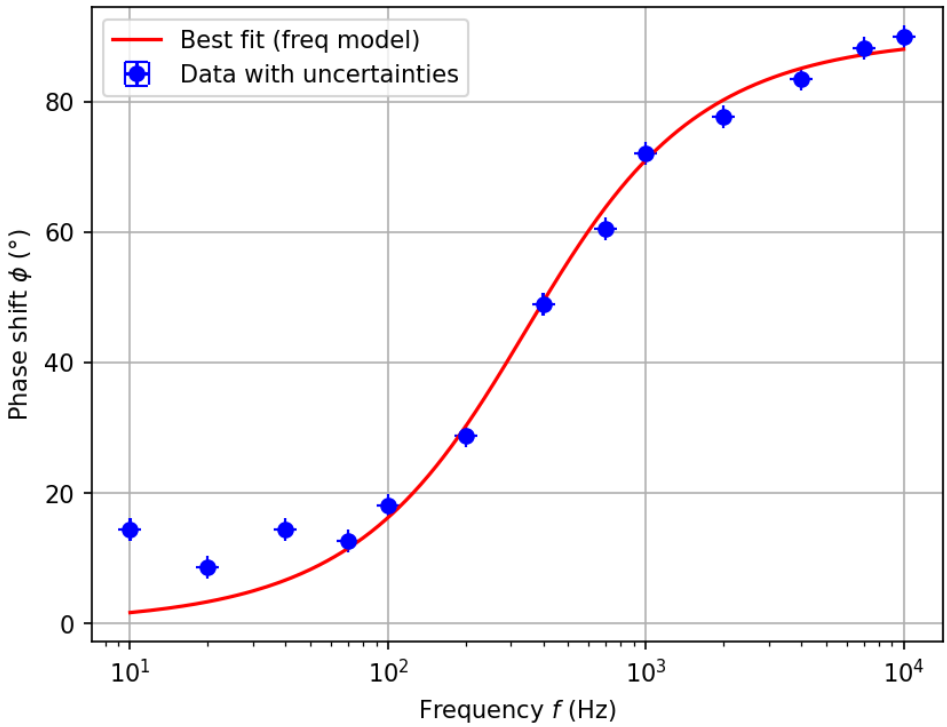


Figure 11: Phase shift *vs* frequency.

f_{char} (Hz)	343,6752 ± 0,0019
χ^2	$3,4 \cdot 10^9$
χ^2_{red}	$2,8 \cdot 10^8$
r^2 value	0,76

Table 5: Values of chi squared χ^2 and reduced chi squared χ^2_{red}; Pearson's correlation coefficient r^2; and the characteristic frequency of the RC circuit obtained experimentally.

As can be seen in figure 11, the phase shift increases with frequency. At low frequencies, the phase difference is close to zero, indicating that V_c and V_e have similar values. As the frequency increases, the capacitor begins to lag more significantly, and the phase shift asymptotically approaches to ninety degrees. This behaviour was expected due to relation 9.

On the other hand, the value of the characteristic frequency of the circuit was experimentally obtained, 343,6752 ± 0,0019 Hz, is near to the one calculated previously (338,63 Hz), although they do differ significantly.

However, it is important to note that both the chi-squared value $\chi^2 = 3.4 \cdot 10^9$ and the reduced chi-squared $\chi^2_{red} = 2.8 \cdot 10^8$ are extremely large. These values indicate a poor fit of the model to the data. In addition, taking into account r^2, which indicates how strongly two parameters are correlated (the closer to ±1, the stronger the relation), we conclude that the value obtained from the graph is satisfactory.

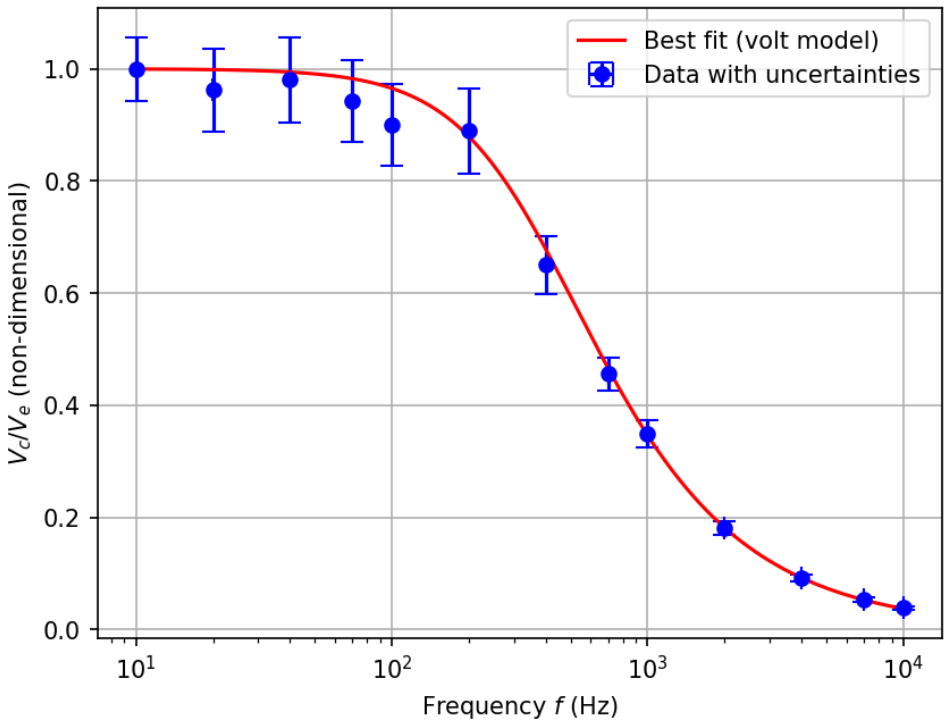


Figure 12: Ratio V_c/V_e *vs* frequency.

f_{char} (Hz)	369 ± 10
χ^2	2,1
χ^2_{red}	0,18
r^2 value	-0,80

Table 6: Values of chi squared χ^2 and reduced chi squared χ^2_{red}; Pearson's correlation coefficient r^2; and the characteristic frequency of the RC circuit obtained experimentally.

As illustrated in figure 12, the ratio V_c/V_e decreases as the frequency increases. At low frequencies, V_c is nearly equal to V_e, resulting in a ratio close to one. However, at higher frequencies, the capacitor becomes less reactive to voltage changes, and the ratio approaches zero. These responses verify equation **??**.

Regarding the statistical parameters, the characteristic frequency obtained from this graph, 369 ± 10 *Hz*, is reasonably close to the theoretical value, although slightly overestimated. The low values of χ^2 and χ^2_{red} suggest that the fit was very good. Additionally, the Pearson's correlation coefficient $r^2 = -0.80$ indicates a strong inverse relationship between f and V_c/V_e, which is consistent with the physical behavior of the system.

3.3 Lissajous figures

In the final section of the experience, the Lissajous figures were studied. Figure 13 contains the images displayed by the oscilloscope when applying different ratios of the two waves' frequencies and phase shifts.

It is important to note that the theoretical values of 0, $\pi/4$ and $\pi/2$ may not be the same as the ones set experimentally because of the internal capacitor of the oscilloscope, which introduces an additional phase shift.

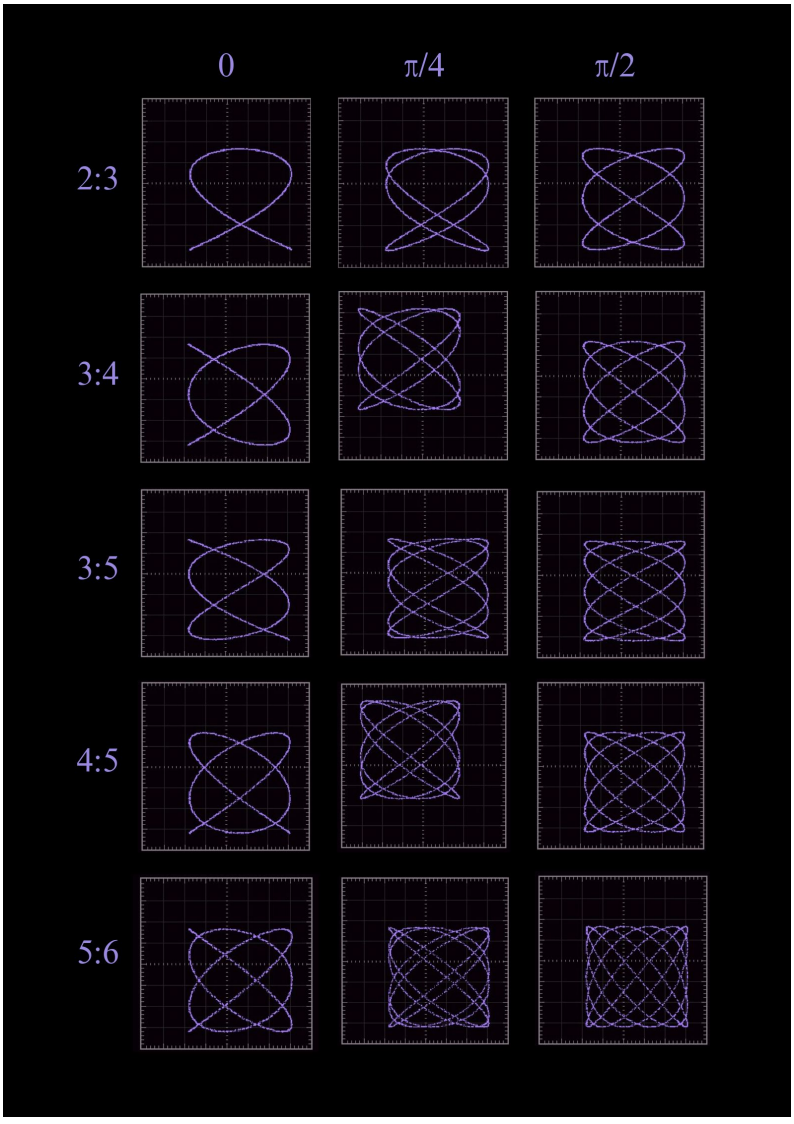


Figure 13: Lissajous obtained using the CHA and CHB signals.

4 Conclusions

This experiment confirmed the oscilloscope's essential role in signal analysis. Using X–T mode, we verified its accuracy in measuring frequency and voltage for common waveforms, with small uncertainties—except for sawtooth signals, where voltage readings were lower than expected.

In the RC circuit analysis, the observed amplitude reduction and phase shift aligned well with theoretical predictions, and the measured characteristic frequency values were close to the theoretical one, though the large chi-squared values in one of the fits suggest some discrepancy (possibly caused by measurement resolution or modeling assumptions).

Lissajous figures in X–Y mode visually illustrated phase differences, showing expected shapes with minor deviations likely due to equipment limitations.

Overall, the results supported theoretical models and highlighted the oscilloscope's value for both practical diagnostics and deeper understanding of circuit behavior.

5 Appendixes

5.1 Calculation of Uncertainties

As we know, type A uncertainty, $u_A(x)$, which is associated to human error, can be calculated with the following equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{12}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations. On the other hand we get type B uncertainty, $u_B(x)$, related to the measuring instrument error from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{13}$$

where δ_x is the resolution of the measuring instrument (δ_L=0,001 m) From them, $u_C(x)$ is expressed as:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{14}$$

Additionally, we can calculate the expanded uncertainty, which is calculated to overestimate the error. To do so, we multiplied $u_C(x)$ by a coverage factor kp that is selected for convenience:

$$U_C(x) = u_C(x)k_p. \tag{15}$$

This factor is derived from the distribution of the t-student, using $n-1$ degrees of freedom and a fixed confidence level. In this case, consistently 95%:

$$kp = 1.959964.$$

For this report, however, we will need to calculate type B uncertainties differently, due to the oscilloscope measurement process. The following proceeding for the uncertainties is provided by the manufacturer:

- Voltage measures: the type B uncertainty for a voltage measure on the oscilloscope divides in: $u_{B,exact}(V_{pp})$ and $u_{B,resol}(V_{pp})$, where V_{pp} is the voltage peak-to-peak of the AC signal. They can be calculated as

$$u_{B,exact}(V_{pp}) = 0.03x + 0.05\Delta x \tag{16}$$

and

$$u_{B,resol}(V_{pp}) = \frac{8\Delta x}{256\sqrt{12}}, \tag{17}$$

where x is the value of the measurement and Δx is the vertical scale of the screen (the number of volts per vertical division).

- Period measures: For the uncertainty of the period, we can get type B uncertainty by

$$u_B(T) = \frac{1}{f_{adq}} + \frac{T}{10^4} + 0.6 \cdot 10^{-9}, \tag{18}$$

where T represents the value of the measurement and f_{adq} the acquisition frequency (times per second the signal is sampled or measured). As there is a constant being added, for this equation there will be always necessary to use SI units.

- Phase measures: type B uncertainties for phase measures are calculated normally, with a resolution of $0.001°$ for the oscilloscope.

On the other hand, for uncertainty type C (u_C), in the case of voltage, although it has two different type B uncertainties, the value of $u_{B,resol}(V_{pp})$ is negligible compared to the value of $u_{B,exact}(V_{pp})$, so we will make the consideration $u_C(V_{pp}) = u_{B,exact}(V_{pp})$.

To continue, to calculate uncertainties in indirect measurements, such as the frequency or the voltage ratio V_c/V_e, as these are values that depend on random variables from direct measurements, we will develop the following equations.

For the uncertainty of the frequency, knowing that $f = \frac{1}{T}$, we get:

$$u_C(f) = \sqrt{\left(\frac{\partial f}{\partial T}\right)^2 (u_C(T))^2} = \frac{u_C(T)}{T^2}. \tag{19}$$

And lastly, for the quantity V_C/V_e:

$$\begin{aligned} u_C(V_C/V_e) &= \sqrt{\left(\frac{\partial V_C/V_e}{\partial V_c}\right)^2 (u_C(V_C))^2 + \left(\frac{\partial V_C/V_e}{\partial V_e}\right)^2 (u_C(V_e))^2} \\ &= \frac{1}{V_e}\sqrt{(u_C(V_C))^2 + \left(\frac{V_C}{V_e} u_C(V_e)\right)^2}. \end{aligned} \tag{20}$$

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